

Database Management System

ER-Diagram Practice

- 1) The university maintains information about its students, courses, faculty members, and tuition payments. Each student is identified by a unique student ID, and the university stores the student's first name, last name, email address, contact number, and the date the student was admitted. The university also determines the student's academic year from the admission date. Students enroll in courses, and each student may take several courses while each course may be taken by many students.

Courses offered by the university are identified by a unique course code, and for each course the university records the course title, the number of credits, and the department to which the course belongs. Courses are taught by faculty members. A faculty member may teach multiple courses, and a course may be taught by one faculty members. Each faculty member is identified by their initials, and the university keeps the faculty member's first name, last name, joining date, and the years of experience from the joining date.

Students also make tuition fee payments. A student may pay tuition in several installments, and each installment has an installment number, payment amount, and payment date. Although the installment number does not uniquely identify a payment across all students, it uniquely identifies an installment for a specific student. The university keeps these payment records to track student fees.

In addition, the university assigns mentors to students. A faculty member may mentor several students, but each student has only one mentor responsible for guiding them academically. Using this information, design the ER diagram for the university system

Diagram:

